(Divisibility and modular Aritmetic)

≡ congruence → Same Remainder

a divides b and ≠0 → Remainder =0

$\frac{b}{a}$

$b = a \times c$ → some integer

notation a|b

Determine wether 3|7 and wether 3|12

$\frac{7}{3}$ → not a integer

$\frac{12}{3}$

$3\overline{)12}\ 4$, $-\frac{12}{0}$

3|12 ✓

$b = a \times c$

$12 = 3 \times 4$

Rule

a|b and a|c then

a | mb + nc where m, n are integers

m, n of our choice

a=10 b=10 c=20

Lets pick m=1, n=3

10|10 → (1) ✓

10|20 → (2)

10 | 10 + 3(20)

10 | 10 + 60

10 | 70 → (7) ✓

Division algo

$a = dq + r$ → remainder

- dividend (no being divided)
- divisor
- quotient

dividend $\frac{5}{4}$ ↓ divisor

r → is never neg and always smaller than divisor.

e.g

$a = 17 \quad d = 5$

$\frac{17}{5}$

$$5 \overline{)17}\ 3$$
$$-15$$
$$2$$

$$17 = 5 \times 3 + 2$$

Extra notations

$$q = a \text{ div } d$$

17 div 5 ↓ $\frac{17}{5}$

$$r = a \bmod d$$ → give remainder

$\frac{3}{-11}$

what the Quotient and remainder when -11 is divided by 3.

$-11/3$

$$3\overline{)-11}$$
$$-9$$
$$-2$$

$$3\overline{)-11}$$
$$-4$$
$$-12$$
$$1$$

→ this no should be smaller

Divisor 3
Remainder $\boxed{0, 1, 2}$

(Modular Arthmatic)

$$a \equiv b \pmod m$$

Dividing both a and b by m, they leave same Remainder hence they are congruent

m/a m/b

$a = 17$ $b = 2$ $m = 5$

$$5\overline{)17}\ 3$$
$$-15$$
$$2$$

$$5\overline{)2}\ 0$$
$$-0$$
$$2$$

$$a \equiv b \pmod 5$$

$a \equiv b \pmod m$ $\quad a = b + km$ for some for some Integer k

$a = 50 \quad b = 2 \quad m = 24$

$\frac{50}{24}$ $\quad 24\overline{)50}\ 2$, -48, 2

$\frac{2}{24}$ $\quad 24\overline{)2}\ 0$, -0, 2

$50 \equiv 2 \pmod{24}$

$50 = 2 + k(24)$

$50 - 2 = 24k$

$\frac{48}{24} = k$

$k = 2$

$\downarrow$

so starting $k = 0 \ldots n$

all these are congruent to

$b = 2 \quad 2, 24, 48, \cdots$

$2 \bmod 48 = 2$

$a \equiv b \pmod m \qquad c \equiv d \pmod m$

$a + c \equiv b + d \pmod m$

$ac \equiv bd \pmod m$

Addition RULE:-

$$(a+b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$$

$$(ad) \bmod m = ((a \bmod m) \cdot (d \bmod m)) \bmod m$$

$(a+b) \bmod m \rightarrow$ rewritten as

$$a + mb$$

$(ad) \bmod m \rightarrow$ rewritten as

$$a + md$$

Primeno:- No: that can only be divided by 1 and itself

A no can be written as a product of prime no by using prime factorisation → Dividing only by primeno

2	100
2	50
5	25
5	5
	1

$$2 \cdot 2 \cdot 5 \cdot 5$$

To check if a no is prime you need to test p.n up to its square root. if any one of them divide then it is composite otherwise prime

$\sqrt{101} = 10.04$

101 is prime

→ 7, 5, 3, 2

11, 13, 17, 19, 23, 29

gcd → two numbers (non zero) and both divide evenly

a = 12 b = 18

12 → 1, 2, 3, 4, 6, 12

18 → 1, 2, 3, 6, 9, 18

gcd (12, 18) = 6

Every integer has atleast 1 as divisor.

17 → 1, 17

22 → 1, 2, 11, 22

gcd (17, 22) = 1

hence 17, 22 is relatively prime as gcd = 1

Set = { 5, 9, 14 }

gcd (5, 9) = 1
gcd (9, 14) = 1
gcd (5, 14) = 1

} called pairwise relatively prime

LCM → multiples of that no

LCM (4, 6)

4, 8, 12, 16, 20, 24

6, 12, 18, 24

LCM → smallest no that appears in both list

LCM (4, 6) = 12

(gcd using prime factorization)

gcd (120, 500)

$120 = 2^3 \cdot 3^1 \cdot 5^1$

$500 = 2^2 \cdot 5^3$

2	120
2	60
2	30
3	15
5	5
	1

2	500
2	250
5	125
5	25
5	5
	1

For each no pick no with lower power.

$2^2 \cdot 3^0 \cdot 5^1$

→ (20)

LCM → pick no with higher power.

$2^3 \cdot 3^1 \cdot 5^3$

---

finding gcd using euclidean Algo (faster way)

gcd (287, 91) → Divide larger by smaller

$\frac{287}{91}$ ≈

$$\begin{array}{r} 3 \\ 91\overline{)287} \\ -\,273 \\ \hline 14 \end{array}$$

$a = b \times q + r$ (a → Larger no, b → Smaller no)

1) gcd (a, b)
2) gcd (b, r)

a becomes b and b becomes r.

$287 = 91 \times 3 + 14$

now doing gcd (b, r)

$\frac{91}{14}$

$$\begin{array}{r} 6 \\ 14\overline{)91} \\ -\,84 \\ \hline 7 \end{array}$$

$a = b \times q + r$

$287 = 91 \times 3 + 14$

$91 = 14 \times 6 + 7$

$14 = 7 \times 2 + 0$

$$\begin{array}{r} 2 \\ 7\overline{)14} \\ -14 \\ \hline 0 \end{array}$$

$\boxed{\gcd(287, 91) = 7}$

e.g gcd (414, 662)

$a = b \times q + r$

$414 = 662 \times 1 + 248$

$b \to a \quad r \to b$

$662 = 248 \times 2 + 166$

$248 = 166 \times 1 + 82$

$166 = 82 \times 2 + 2$

$82 = 2 \times 41 + 0$

$\gcd(414, 662) = 2$

$414 \overline{)662}\ 1$, -414, 248

$248 \overline{)662}\ 2$, -496, 166

$166 \overline{)248}\ 1$, -166, 82

$82 \overline{)166}\ 2$, -164, 2

$2 \overline{)82}\ 41$, -82, 0

gcd (414, 662)

$a = bq + r$

$662 = 414 \times 1 + 248$

$414 = 248 \times 1 + 166$

$248 = 166 \times 1 + 82$

$166 = 82 \times 2 + 2$

$82 = 41 \times 2 + 0$

(gcd as linear combinations)

$$\gcd(a,b) = sa + tb$$

$a = 6 \quad b = 14$

$\gcd(6,14) = 2$

$6 = 1, 2, 3, 6$

$14 = 1, 2, 7, 14$

writing in euclidean form

$$6 \overline{)14} \quad 2, \quad -12, \quad 2$$

$a = bq + r$

$14 = 6 \times 2 + 2$

a is always the largest no.

making Rem the subject

$14 - 6(2) = 2$

$2 = 14 \times 1 - 6 \times 2$ (changes sign of entire term)

$2 = 1 \times 14 - 2 \times 6$

$sa + tb$

$s = 1, \quad t = -2$

$ac \equiv bc \pmod{m}$ and $\gcd(c,m) = 1$ then

$a \equiv b \pmod{m}$

↓

multiplying a and b by the same no and we can only cancel c out if gcd of c and m = 1.

$16 \equiv 4 \pmod 3$

$4 \equiv 1 \pmod 3$ ✓

$\gcd(4,3)$

$= (1)$ —

$3 \to 1, 3$

$4 \to 1, 4$

Linear congruency:. $ax = b \pmod m$ finding value of x

Modular inverse $a\bar{a} \equiv 1 \pmod m \rightarrow x \equiv a^{-1} b \pmod m$

When inverse exist :. $\gcd(a,m) = 1$

e.g

$3x \equiv 4 \pmod 7$

$3 = 1, 3$

$7 = 1, 7$

$\gcd(a,m) = \gcd(3,7) \equiv 1$ —

inverse exist

finding inverse (find a^{-1})

$3 \times a^{-1} \equiv 1 \pmod 7$

$3x \equiv 4 \pmod 7$

$a = bq + r$

$\frac{7}{3} = \frac{b}{a}$

$3 \overline{)7}^{2}$

-6

(1)

$7 = 3(2) + 1$

$7 - 3 \times 2 = 1$

$1 = 7 - 3 \times 2$

$1 = 1 \times 7 - 2 \times 3$

$\text{Rem} = tb + sa$

$s = -2$

$t = 1$

$1 \equiv 1 \times 7 - 2 \times 3 \pmod 7$

$1 \times 7 \pmod 7$

$3n = 4 \pmod 7$

$\gcd(3, 7) = 1$

$7 = 3 \times 2 + 1$ → proves

$3 = 1 \times 3 + 0$

Solving Backwards :: Making n the Subject

$1 = 7 - 3 \times 2$

$[1 = -2 \times 3 + 1 \times 7] \pmod 7$ → $7 \pmod 7 \equiv 0$

(a) (b)

↓

Checking $-6 \bmod 7 = 1$

$$7 \overline{)-6} \quad -1 \quad \pm 7 \quad (1)$$

Hence -2 is the inverse

This also satisfys the LHS 1. remainder.

$3x \equiv 4 \pmod 7$ multiplying both sy -2

$-2 \times 3x \equiv -2 \times 4 \pmod 7$

Reducing

$-6x \pmod 7$

$-6 \pmod 7$

$$7 \overline{)-6} \quad -1 \quad \pm 7 \quad (1)$$

$= (1)$ so

$-8 \bmod 7$

$\equiv 6$

again multply by x

$$7 \overline{)-8} \quad -2 \quad \pm 14 \quad (6)$$

Rem: $1x$ Shows that -2 was indeed a inverse.

$$\boxed{x \equiv 6 \pmod 7}$$

$3n \equiv 4 \pmod 7$ $\quad n = 6$

18 mod 7 $\quad$ 4 mod 7

$$7 \overline{)18}\ 2 \qquad -14 \qquad (4)$$

$$7 \overline{)4}\ 0 \qquad -0 \qquad (4)$$

---

find Inverse of $101\bar{a} \equiv 1 \pmod{4620}$

First using ecudlian theorem to check if gcd is 1

$a = b \times q + r$ $\quad$ so basically we need to find $\bar{a}$

$a\bar{a} \equiv 1 \pmod m$

gcd(101, 4620)

$$101 \overline{)4620}\ 45 \qquad -4545 \qquad 75$$

$4620 = 101 \times 45 + 75$

$$75 \overline{)101}\ 1 \qquad -75 \qquad 26$$

$101 = 75 \times 1 + 26$

$$26 \overline{)75}\ 2 \qquad -52 \qquad 23$$

$75 = 26 \times 2 + 23$

gcd: 1 hence Inverse exists

$26 = 23 \times 1 + 3$

$23 = 7 \cdot 3 + 2$

$3 = 1 \cdot 2 + (1)$

$2 = 1 \cdot 2 + 0$

Now Reversing :(Making Remainder the Subject)

$1 = 3 - 1 \cdot 2$ ← Rem

$2 = 23 - 7 \cdot 3$

$= 3 - 1(23 - 7 \cdot 3)$

$= 3 - 23 + 21$

$= 3 - 23 + 7(26 - 1 \cdot 23)$

$= 3 - 23 + 182 - 7(23)$

$= 3 - (75 - 2 \cdot 26) + 182 - 7(75 - 2 \cdot 26)$

$= 3 - 75 + 2 \cdot 26 + 182 - 7(75 - 2 \cdot 26)$

$\text{lcm}(n, 0) = 0$

$\gcd(0, n) = n$

$\gcd(2^a - 1, 2^b - 1)$

$= 2^{\gcd(a,b)} - 1$

$n \equiv 2 \pmod 3$ $\quad n \equiv 3 \pmod 5$ $\quad n \equiv 4 \pmod 7$

$M \to 3 \times 5 \times 7 \to 105$

$m_1 = \frac{105}{3} = 35$

$m_2 = \frac{105}{5} = 21$

$m_3 = \frac{105}{7} = 15$

y_1 :- inverse of $M_1 \pmod 3$

$35 y_1 \equiv 1 \pmod 3$

$(a = bq + r)$

$35 = 3 \times 11 + 2$

$\boxed{1 = 3 - 2 \times 1}$ $\qquad 2 = 35 - 3 \times 11$

$1 = 3 - (35 - 3 \times 11) \times 1$

$1 = 3 - 35 + 3 \times 11$

$1 = -35 + 3 \times 11 + 3$

$1 = -35 + 3(11 + 1)$

$1 = -35 + 3 \times 12$

$y_1 = -2$

$-1 \pmod 3$

$1 = -1 \times 35 + 3 \times 12$

So $y_1 = -1$ first add 3 to make image positive $-1+3 \Rightarrow 2$

$a^{p-1} \equiv 1 \pmod{p}$

$7^{222} \bmod 11$

$7^{10} \equiv 1 \pmod{11}$

$\equiv 1 \pmod{11}$

$\equiv 1 \pmod{11}$

$7^{10} = 1 \pmod{11}$